

Task 1: Create two new users on your Linux system named 'musicfan' and 'moviebuff', then create a new group called 'playlist' and add both users to this group.

Ans :-

Create the users

useradd musicfan

passwd musicfan

useradd moviebuff

passwd moviebuff

Create the group

groupadd playlist

Add both users to the group

usermod -aG playlist musicfan

usermod -aG playlist moviebuff

Verify

id musicfan

id moviebuff

Expected output should include:

playlist

**Task 2: Inside your home directory, create a folder called 'spotify_shared'. Set its permissions so only users in the 'playlist' group can read and write files inside it, but others cannot access it.

Hint: Use chown and chmod with the group ownership and permission bits.**

Ans :-

Create the folder inside your home directory (assuming you are root):

mkdir ~/spotify_shared

Change the group ownership:

chown root:playlist ~/spotify_shared

Give permissions:

```
chmod 770 ~/spotify_shared
```

Verify:

```
ls -ld ~/spotify_shared
```

Example output:

```
drwxrwx--- 2 root playlist ...
```

Explanation

- Owner (root): Read, Write, Execute
- Group (playlist): Read, Write, Execute
- Others: No permissions

Only members of the **playlist** group can access the directory.

Task 3: Place a text file named 'top_songs.txt' inside 'spotify_shared'. Change its ownership so that 'musicfan' is the owner and 'playlist' is the group, then set permissions so only the owner can edit, group members can read, and others have no access.

Ans :-

Create the file:

```
touch ~/spotify_shared/top_songs.txt
```

Add some content:

```
echo "Shape of You" > ~/spotify_shared/top_songs.txt
```

Change ownership:

```
chown musicfan:playlist ~/spotify_shared/top_songs.txt
```

Set permissions:

```
chmod 640 ~/spotify_shared/top_songs.txt
```

Verify:

```
ls -l ~/spotify_shared/top_songs.txt
```

Example output:

```
-rw-r----- 1 musicfan playlist 13 Jul 31 top_songs.txt
```

Permission Breakdown

640

Owner (musicfan)

- Read
- Write

Group (playlist)

- Read
- Write

Others

- No access

Task 4: Simulate a scenario where 'moviebuff' tries to access 'top_songs.txt'. Switch to the 'moviebuff' user and attempt to open and edit the file. Describe what happens and explain why based on the permissions you set.

Ans :-

Switch user:

```
su - moviebuff
```

Read the file:

```
cat /root/spotify_shared/top_songs.txt
```

If the folder is inside /root, this will fail because /root is typically 700 and inaccessible to other users, even if spotify_shared itself has group permissions.

A better location for testing is to place the shared directory somewhere accessible, such as:

```
mkdir /shared
```

```
mv /root/spotify_shared /shared/
```

or create it directly under /home:

```
mkdir /home/spotify_shared
```

Then:

```
cat /shared/spotify_shared/top_songs.txt
```

Expected output:

Shape of You

Try editing:

```
echo "Believer" >> /shared/spotify_shared/top_songs.txt
```

Expected output:

```
bash: /shared/spotify_shared/top_songs.txt: Permission denied
```

Why This Happens

moviebuff is a member of the **playlist** group.

The file permissions are:

```
-rw-r-----
```

which means:

- Owner (musicfan) → Read and Write
- Group (playlist) → Read only
- Others → No access

Therefore:

- moviebuff can open and read the file because of group read permission.
- moviebuff cannot edit or append to the file because the group does not have write permission.